



pabilities. Unlike GPUs that sequentially load entire networks from external memory, memristors replicate features of both artificial and biological neural networks, eliminating the need for external memory and saving energy. Using YBCO superconductor and various oxides, the team achieved time constants between 159 and 278 nanoseconds. The memristor network could learn and recognise digits zero through nine, identifying numbers before audio input completion.

Enhancing object-tracking abilities in self-driving cars

Researchers at the University of Toronto Institute for Aerospace Studies (UTIAS) have introduced two innovations to enhance autonomous vehicle safety by improving multi-object tracking, essential for monitoring movements in crowded areas. These



A visualisation of a nuScenes dataset used by the researchers. The image is a mosaic of the six different camera views around the car with the object bounding boxes rendered over top of the images (Image Courtesy of the Toronto Robotics and AI Laboratory)

systems use 2D cameras and 3D lidar, predicting object movements at ten times per second. Their 2D tracking-by-detection methods use probabilistic object detection to quantify uncertainties, managing errors from low-light or occlusion. This research, focusing on visual anomaly detection for Canadarm3 in the Gateway lunar outpost project, aims to develop AI methods that understand object persistence and reason about new or unexpected encounters.

High performance transistors on 200mm wafer

Researchers at Samsung Advanced Institute of Technology (SAIT) and Seoul National University successfully integrated MoS₂ transistors on a 200mm wafer, showcasing their potential for smaller, flexible devices. They utilised metal-organic chemical vapour deposition (MOCVD) to fabricate large-scale arrays of MoS₂ FETs, enhancing carrier mobility by eliminating the Schottky barrier at the MoS₂-metal interface. This method, compatible with current manufacturing processes, achieved over 99.9% yield.

Desalination method cuts energy use by 80%

Researchers from the Australian National University have developed a new desalination method reducing energy use by 80%. With freshwater scarcity worsening globally,



Shuqi Xu (left) and Prof. Juan Felipe Torres with their first prototype of thermodiffusive desalination (Credit: ANU College of Engineering, Computing & Cybernetics)

regions like Israel, Mexico City, India, and southern California increasingly depend on desalination. Desalination currently serves 300 million people and consumes 3kWh/m³ of energy, contributing significantly to water

provision costs. Traditional methods like reverse osmosis harm marine life and are maintenance intensive. The new method uses thermodiffusion, moving salt to colder areas, producing low salinity water efficiently without membranes or phase changes.

Turning up the heat on next-generation semiconductors

Researchers from MIT have adopted gallium nitride (GaN) for high-temperature applications, such as exploring Venus, due to its ability to withstand over 500°C. GaN, used in phone chargers and cell towers, is now being tested beyond 300°C, the limit for traditional silicon electronics. Using MIT.nano facilities, they created GaN devices with resistors to measure material and contact resistance. Two techniques applied ohmic contacts: metal deposition with annealing and replacing portions with highly doped GaN. Short-term tests at 500°C and long-term tests for 72 hours showed stable contact resistance initially, but degradation began after 48 hours.

Lightweight night vision for everyday use

Researchers at TMOS, the ARC Center of Excellence for Transformative Meta-Optical Systems, have developed an ultra-thin infrared filter that could be integrated into regular eyewear, enabling simultaneous viewing of infrared and visible light. This innovation promises to improve safety in nighttime driving, evening walks, and low-light work environments by eliminating the need for bulky night vision systems. The new metasurface-based technology operates at room temperature, bypassing the need for cryogenic cooling, and merges infrared and visible light into a single coherent image.